

Logic Gates — Paper A

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. The mark for each question is in square brackets on the right.
- No calculator is needed anywhere on this paper.
- Truth-table rows go in **binary counting order**: 00, 01, 10, 11, and for three inputs 000, 001, 010, 011, 100, 101, 110, 111.
- Write expressions with AND, OR, NOT, NAND and NOR, and put **brackets wherever two different gates meet**. NOT applies only to the thing straight after it, so NOT A AND B means (NOT A) AND B.
- A reminder of the five gates is printed below. You may use it on every question.

REMINDER • THE FIVE GATES • YOU MAY USE THIS ON EVERY QUESTION

- AND**1 only when **every** input is 1.
- OR**1 when **at least one** input is 1. Both inputs being 1 still gives 1.
- NOT**One input. The output is the opposite of it.
- NAND**AND, then flip. 0 only when every input is 1; 1 on every other row.
- NOR**OR, then flip. 1 only when every input is 0; 0 on every other row.

SECTION 1 – GATES AND TABLES

1. A circuit has **three** inputs, A, B and C.

(a) How many rows does its truth table need? [1]

(b) Write out every row of the table below, in counting order, starting at 000. [2]
(There is no output column yet.)

A	B	C

2. Complete the truth table for an **AND** gate and for an **OR** gate with the same two inputs. [2]

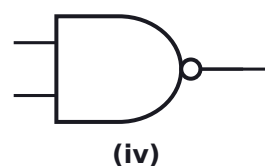
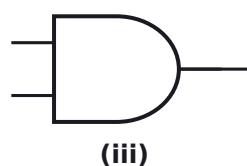
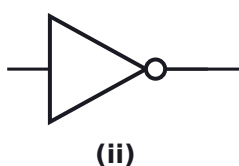
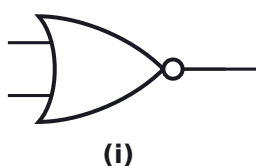
A	B	AND	OR
0	0		
0	1		
1	0		
1	1		

3. Complete the truth table for a **NAND** gate and for a **NOR** gate with the same two inputs. [2]

A	B	NAND	NOR
0	0		
0	1		
1	0		
1	1		

Do the plain gate first in your head, then flip every output.

4. Name each of these four gate symbols. Write the name in the box under its number. [4]

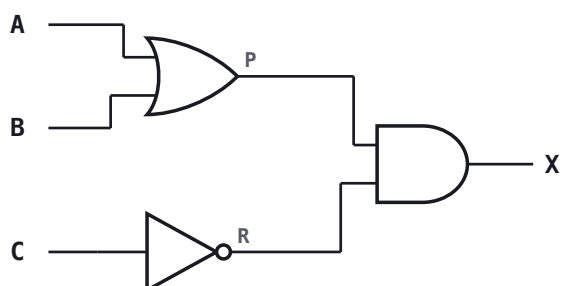


(i)	(ii)	(iii)	(iv)

Look for a small circle first, then at the shape of the body. A circle on its own means “flip”.

SECTION 2 – READING CIRCUITS

5. Study the circuit. The first two gates give the outputs P and R, and the last gate gives X.



- (a) Write the expression for X. [1]

(b) Complete the trace table. Fill in one whole column at a time, from left to right. [3]

A	B	C	P	R	X
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

6. NOT applies only to the thing straight after it. Complete the table for these two expressions, which look alike:

$$X = \text{NOT } A \text{ OR } B \quad Y = \text{NOT } (A \text{ OR } B)$$

(a) Complete the table. [2]

A	B	X	Y
0	0		
0	1		
1	0		
1	1		

(b) One of X and Y gives exactly the same output column as a single gate. Name the gate, and say whether it is X or Y. [1]

SECTION 3 – FROM WORDS TO CIRCUITS

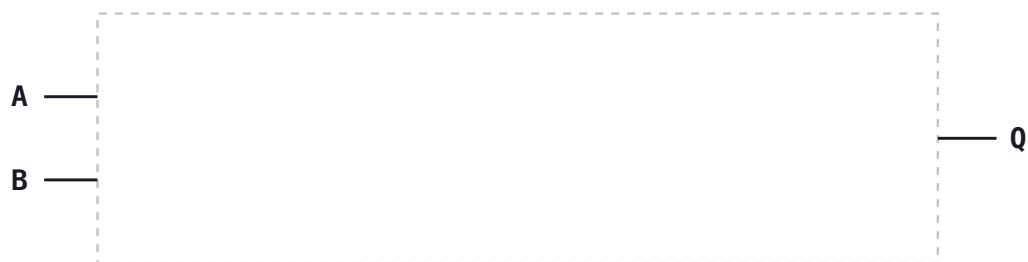
7. A pump fills the overhead tank of an apartment block in Bengaluru. The pump motor, Q, must run only when the tank is **not** full **and** the sump (the underground tank) has water in it. **A = 1** when the overhead tank is full. **B = 1** when the sump has water.

(a) Write an expression for Q. [1]

(b) Complete the truth table for Q. [2]

A	B	Q
0	0	
0	1	
1	0	
1	1	

- (c) Draw the circuit in the box, using the standard gate symbols. [2]



8. An auto-rickshaw has a “FOR HIRE” lamp, H. $M = 1$ when the fare meter is running. $P = 1$ when a passenger is seated. The lamp must be on only when the meter is off **and** nobody is seated.

- (a) Write an expression for H. [1]

- (b) Complete the truth table for H. [2]

M	P	H
0	0	
0	1	
1	0	
1	1	

- (c) Which single gate gives this same table? [1]

9. In this circuit the two inputs of one **NAND** gate are joined together, so both receive the same input A.



- (a) Complete the truth table for Z. [2]

A	Z
0	
1	

- (b) Which single gate does this circuit behave as? [1]

End of paper. Check every truth table has all its rows in counting order, and that every expression has brackets wherever two different gates meet.